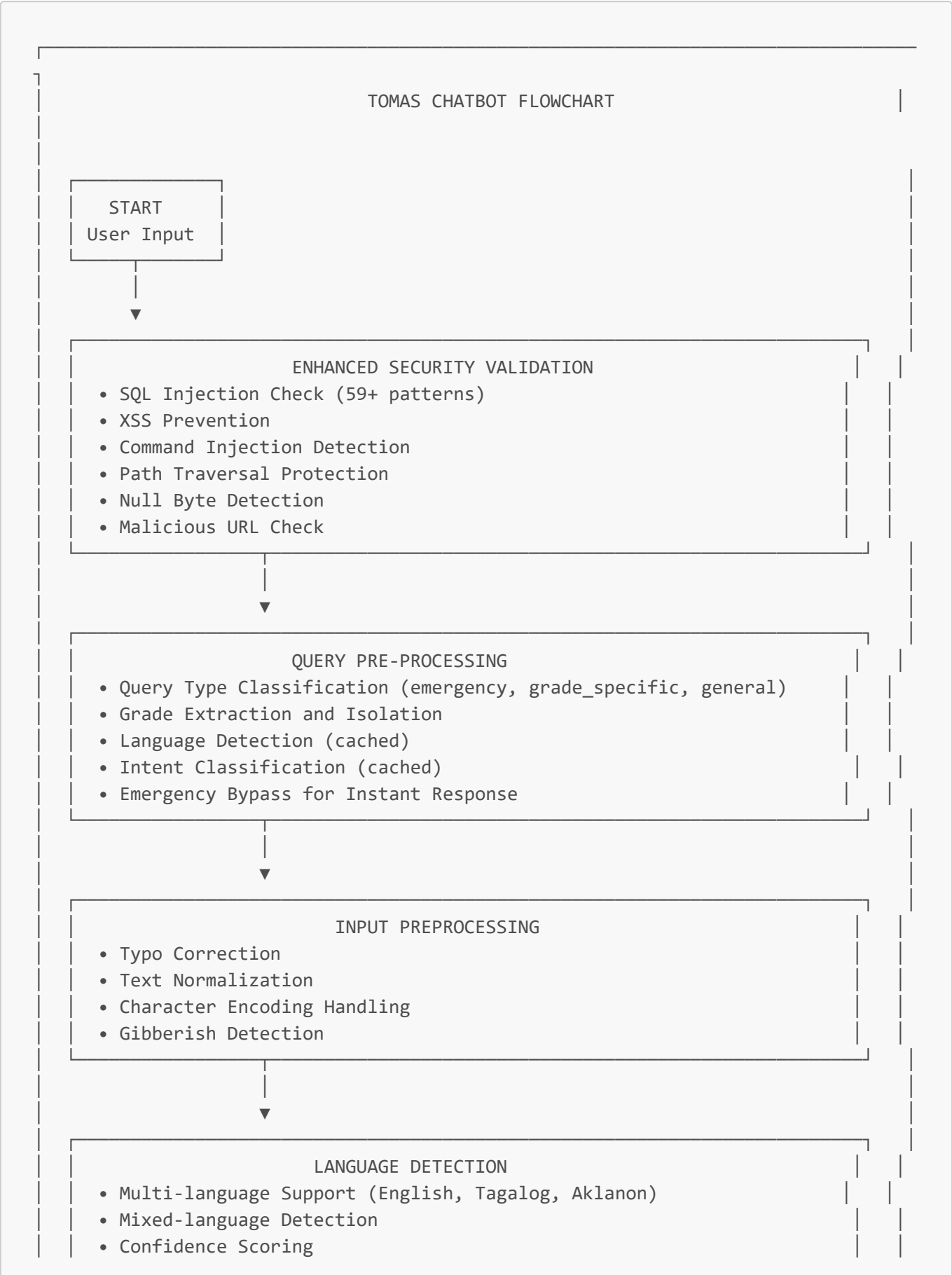
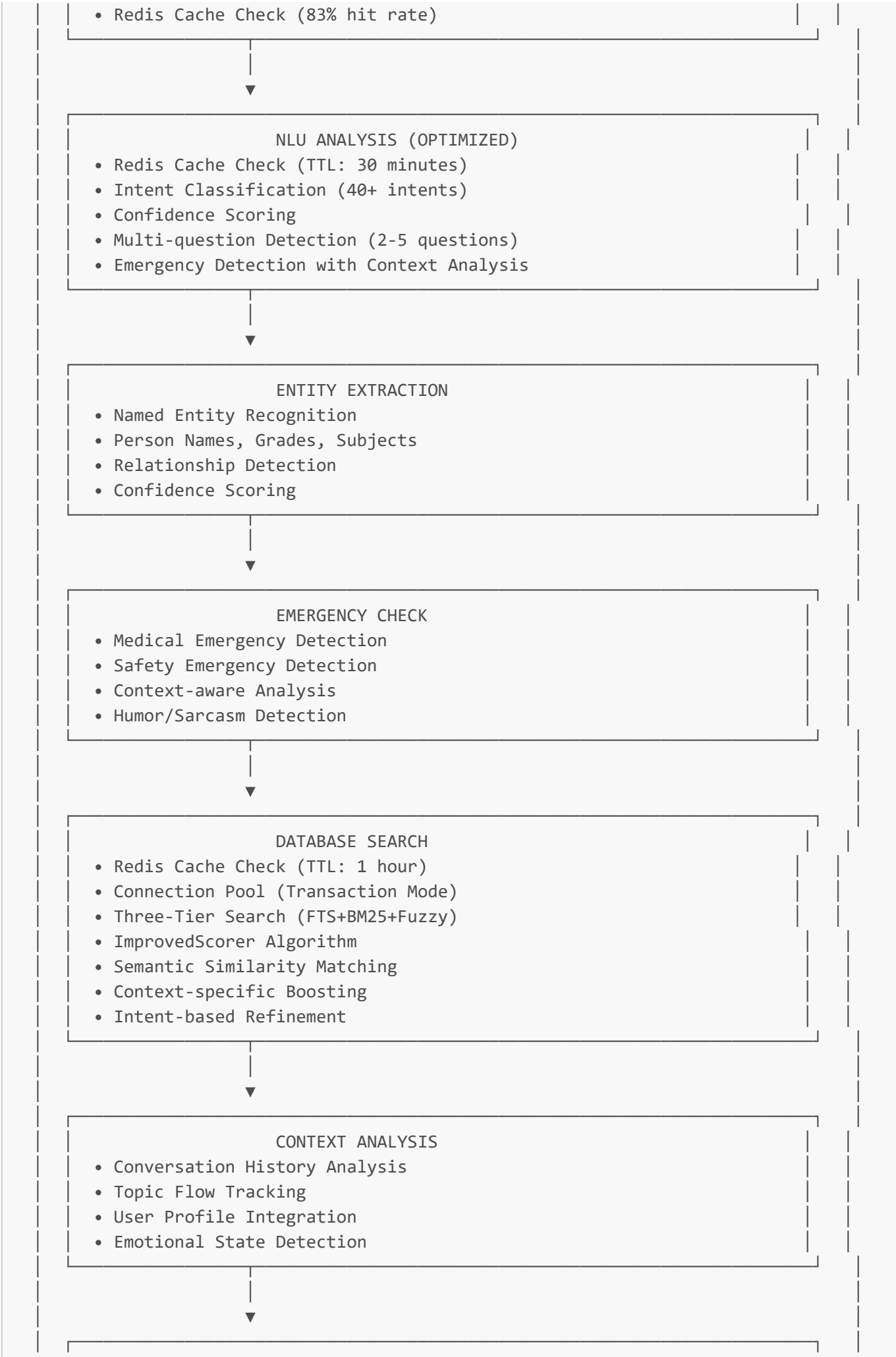
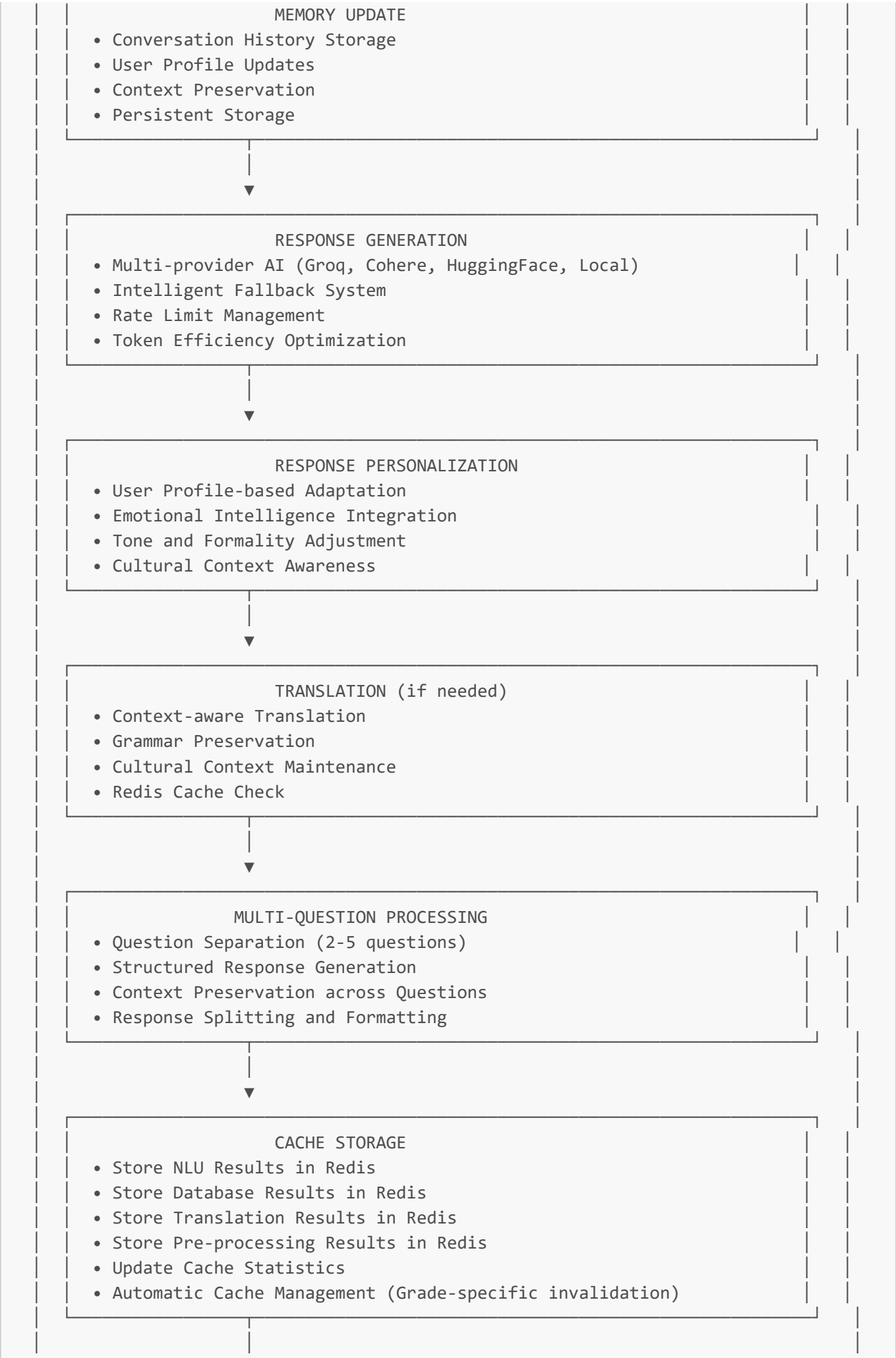


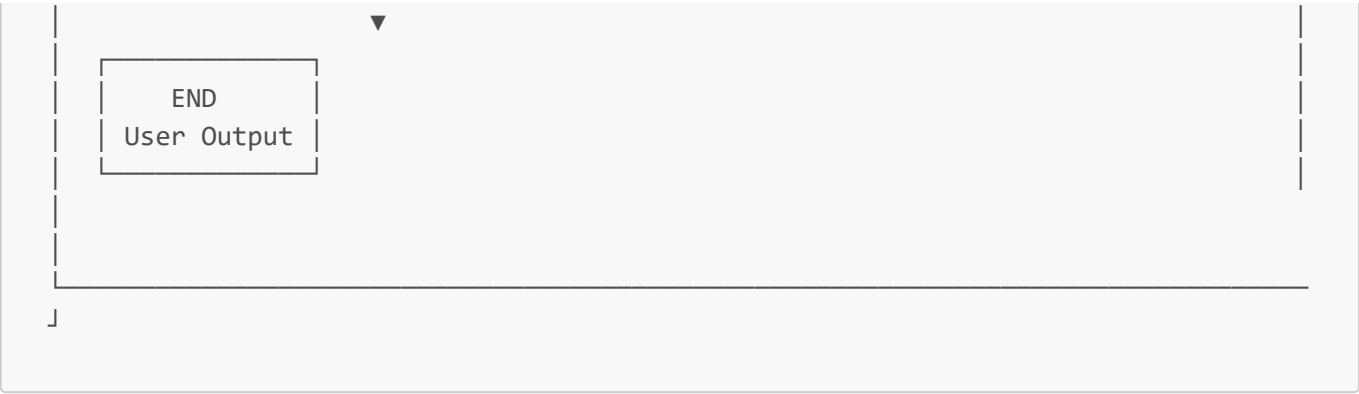
TOMAS Chatbot Processing Flowchart

Main Processing Flow

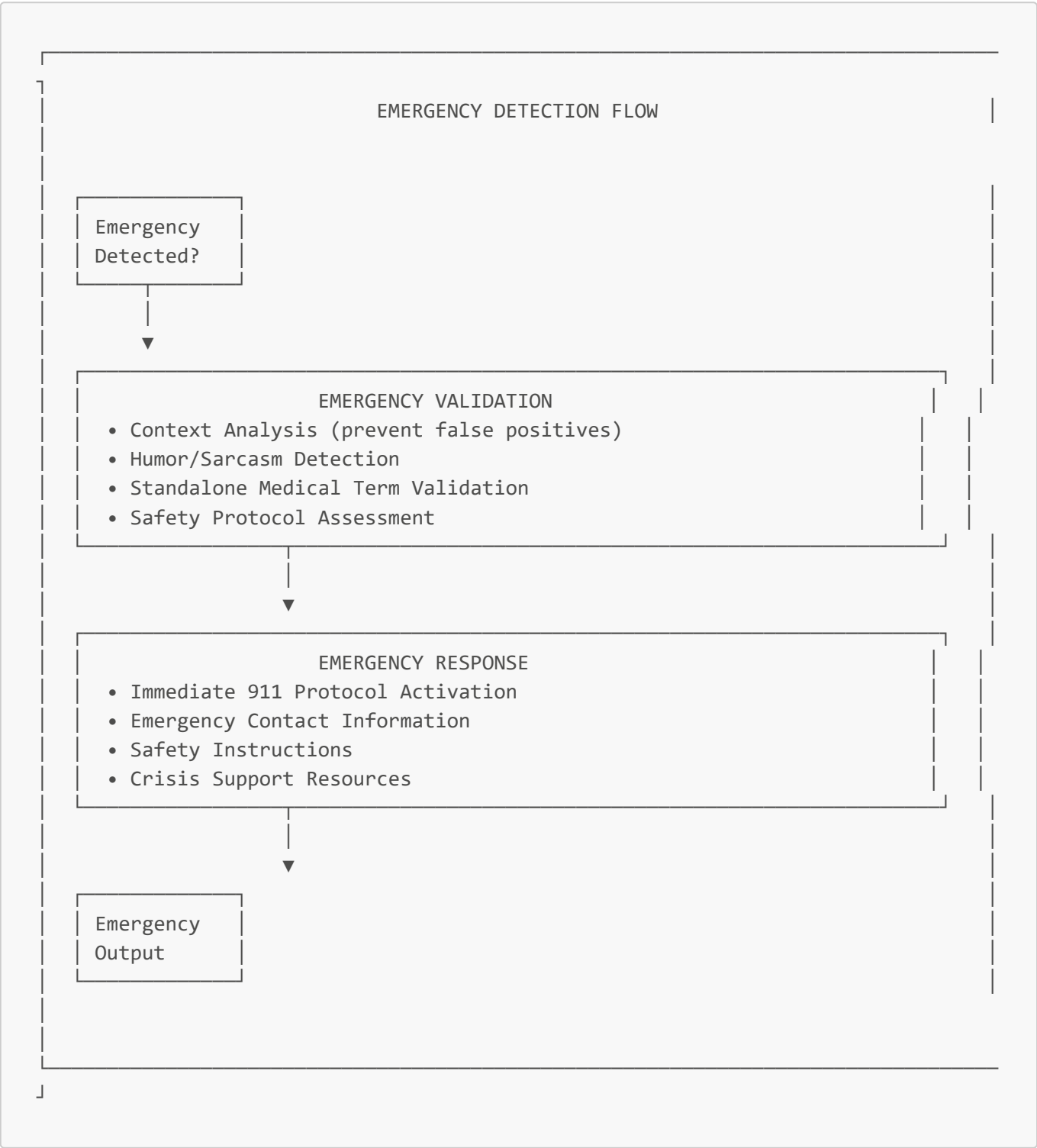




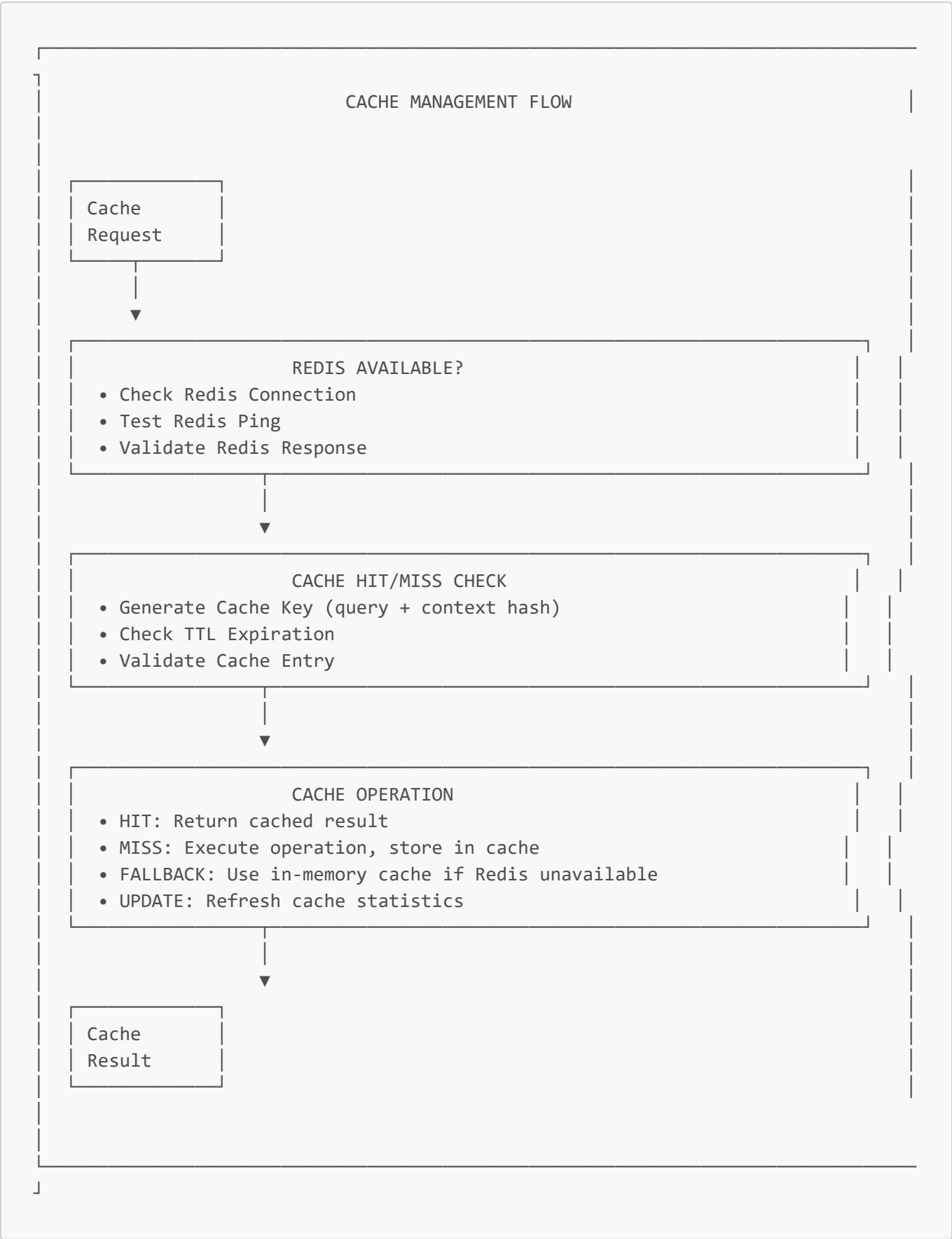




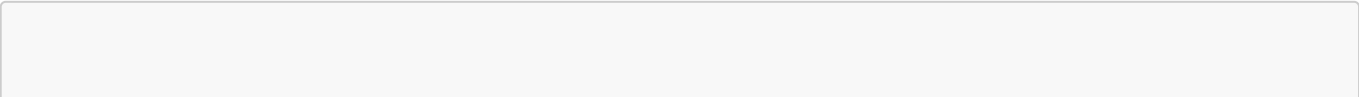
Emergency Detection Flow

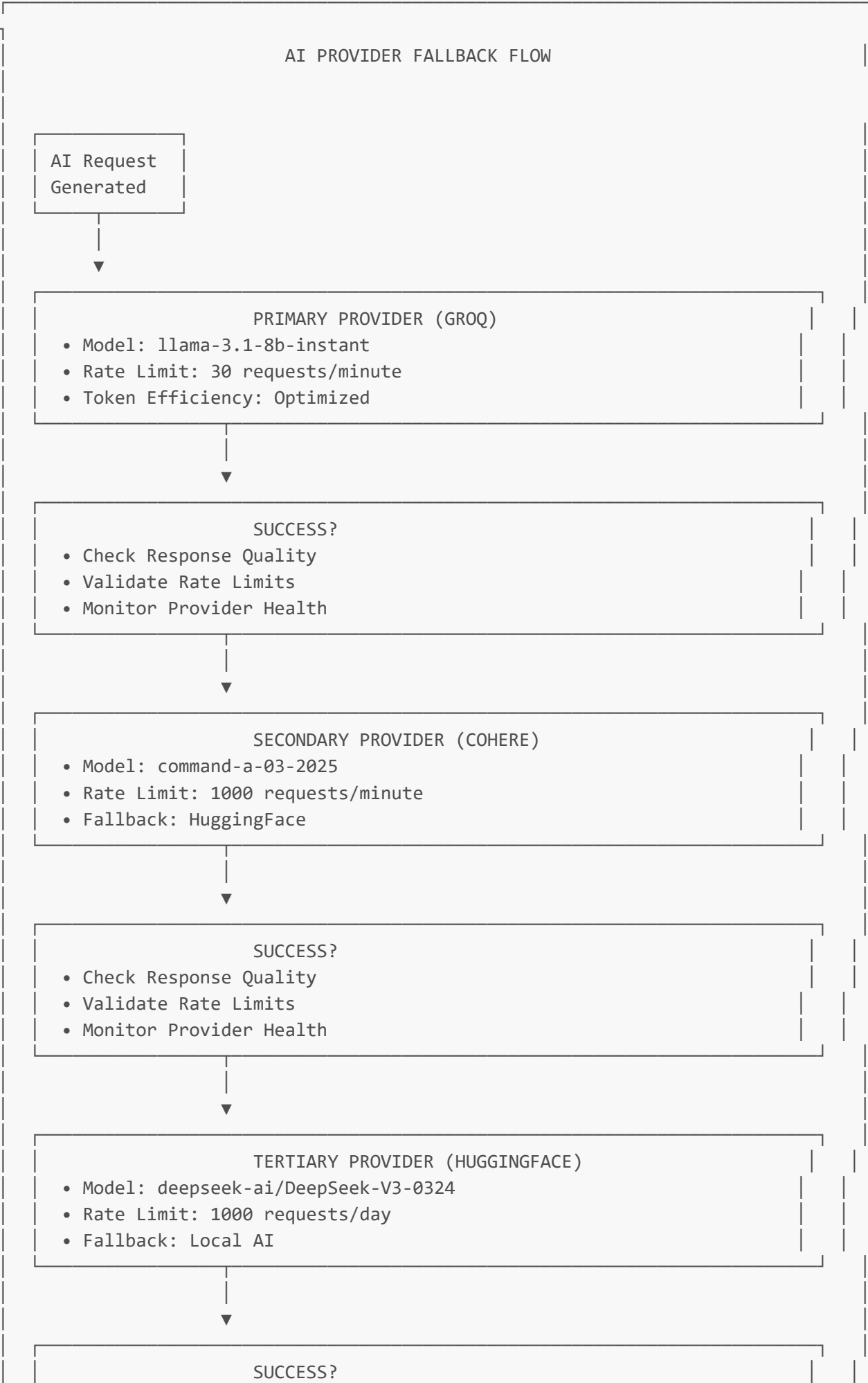


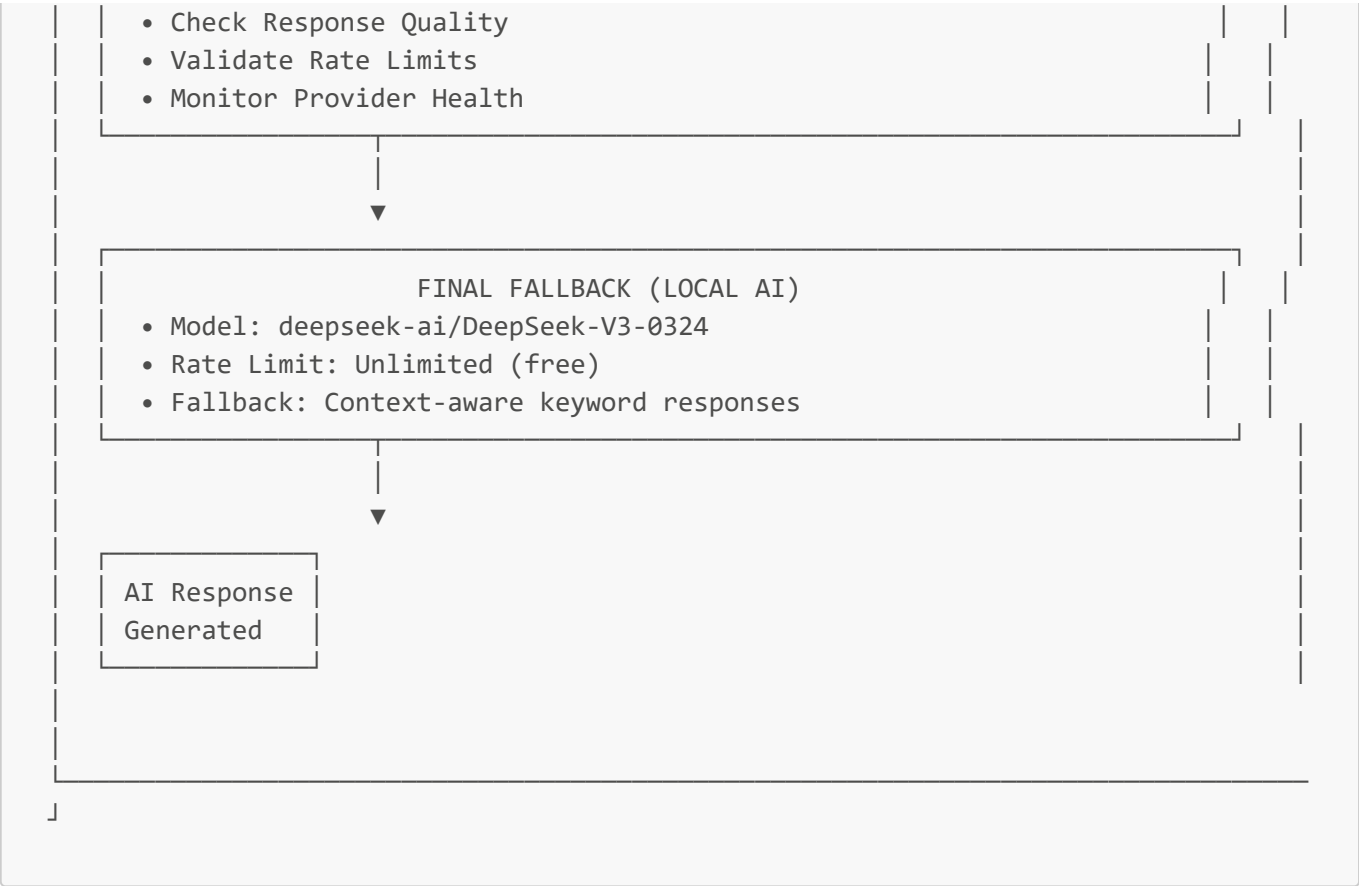
Cache Management Flow



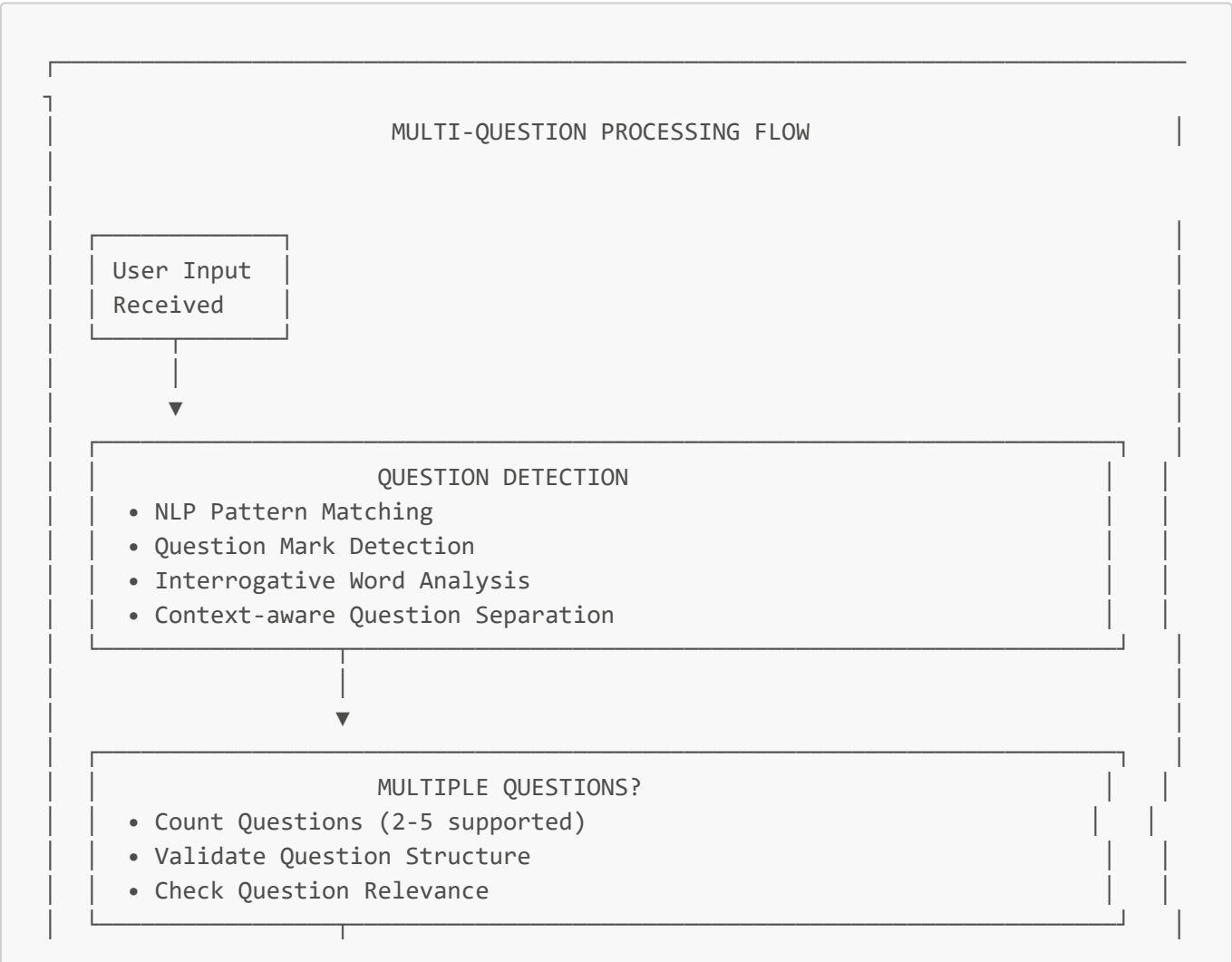
AI Provider Fallback Flow

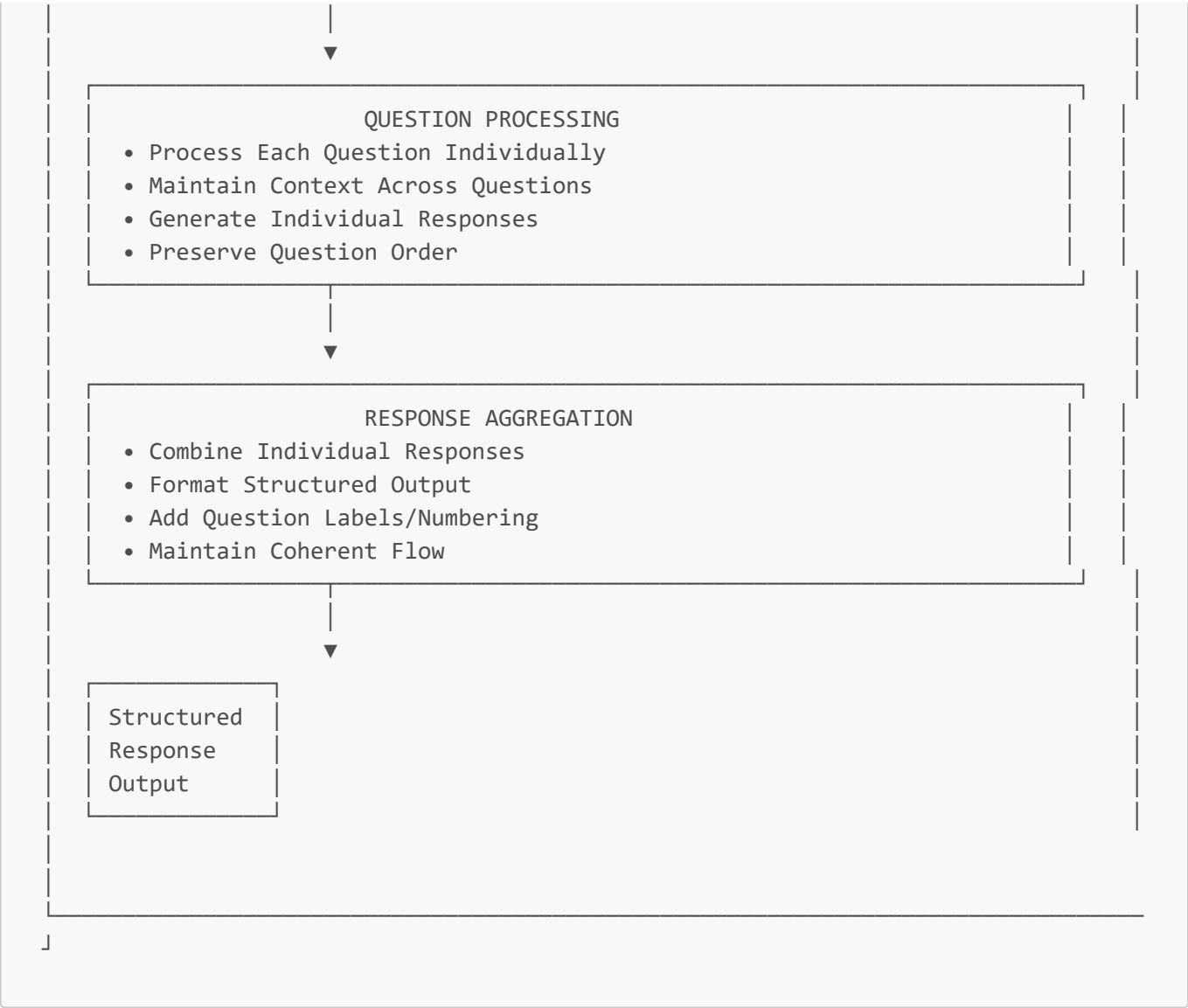




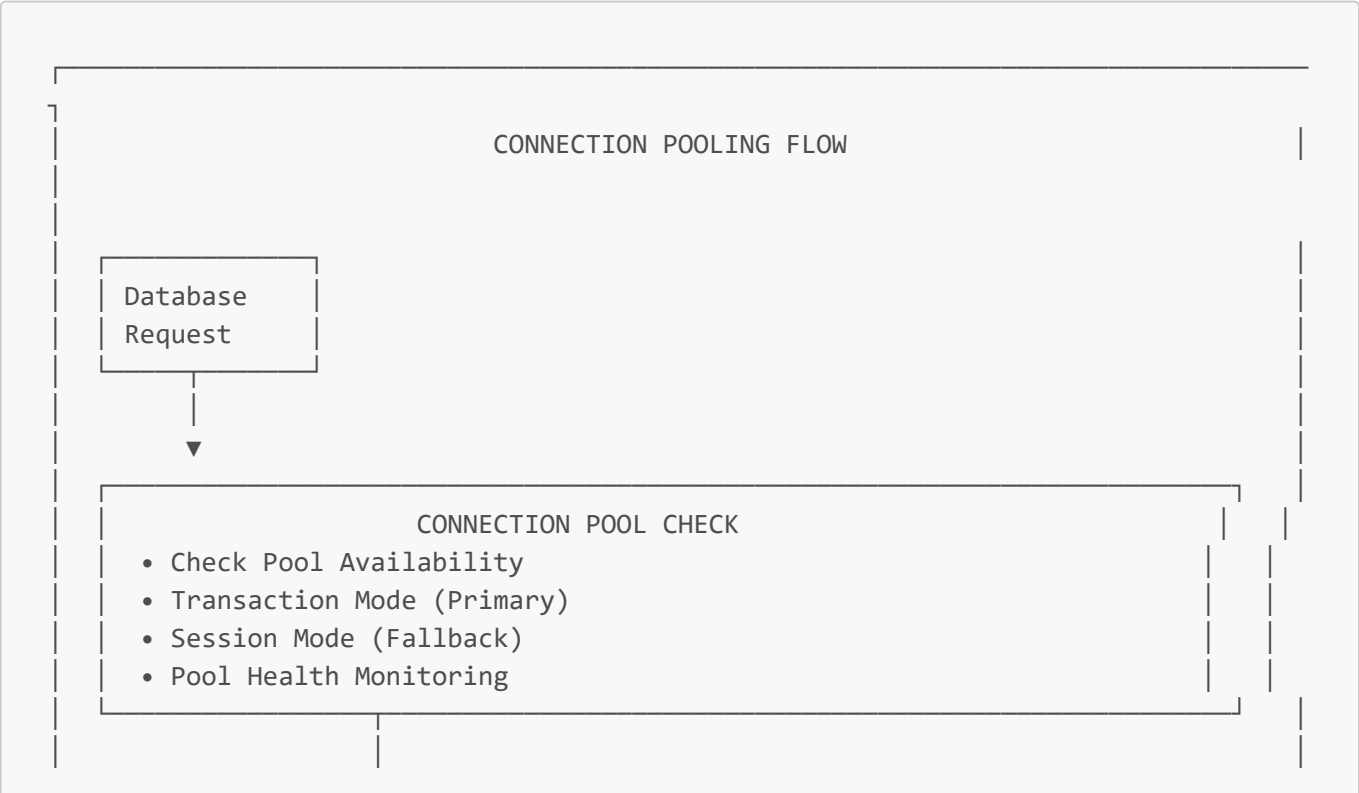


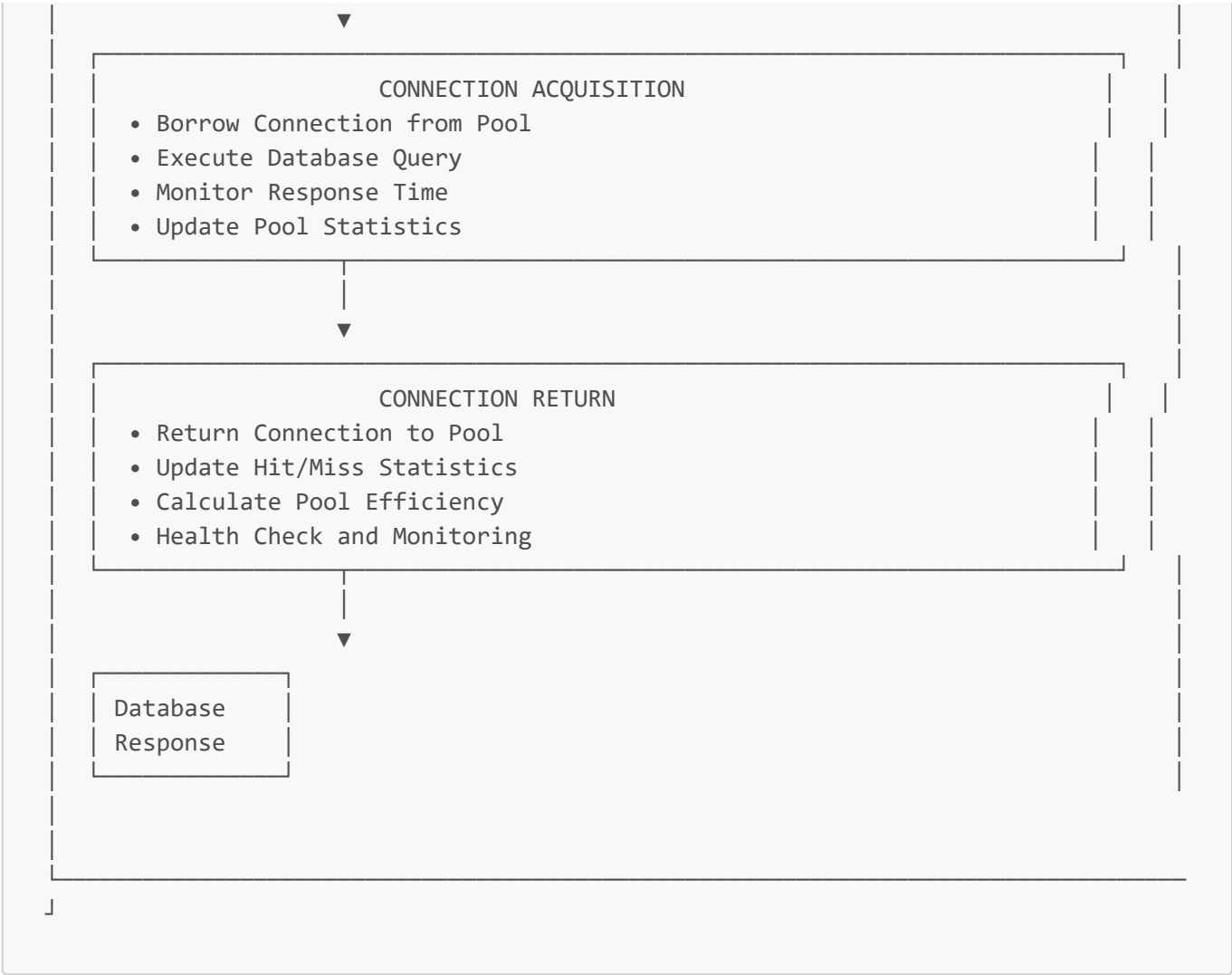
Multi-Question Processing Flow



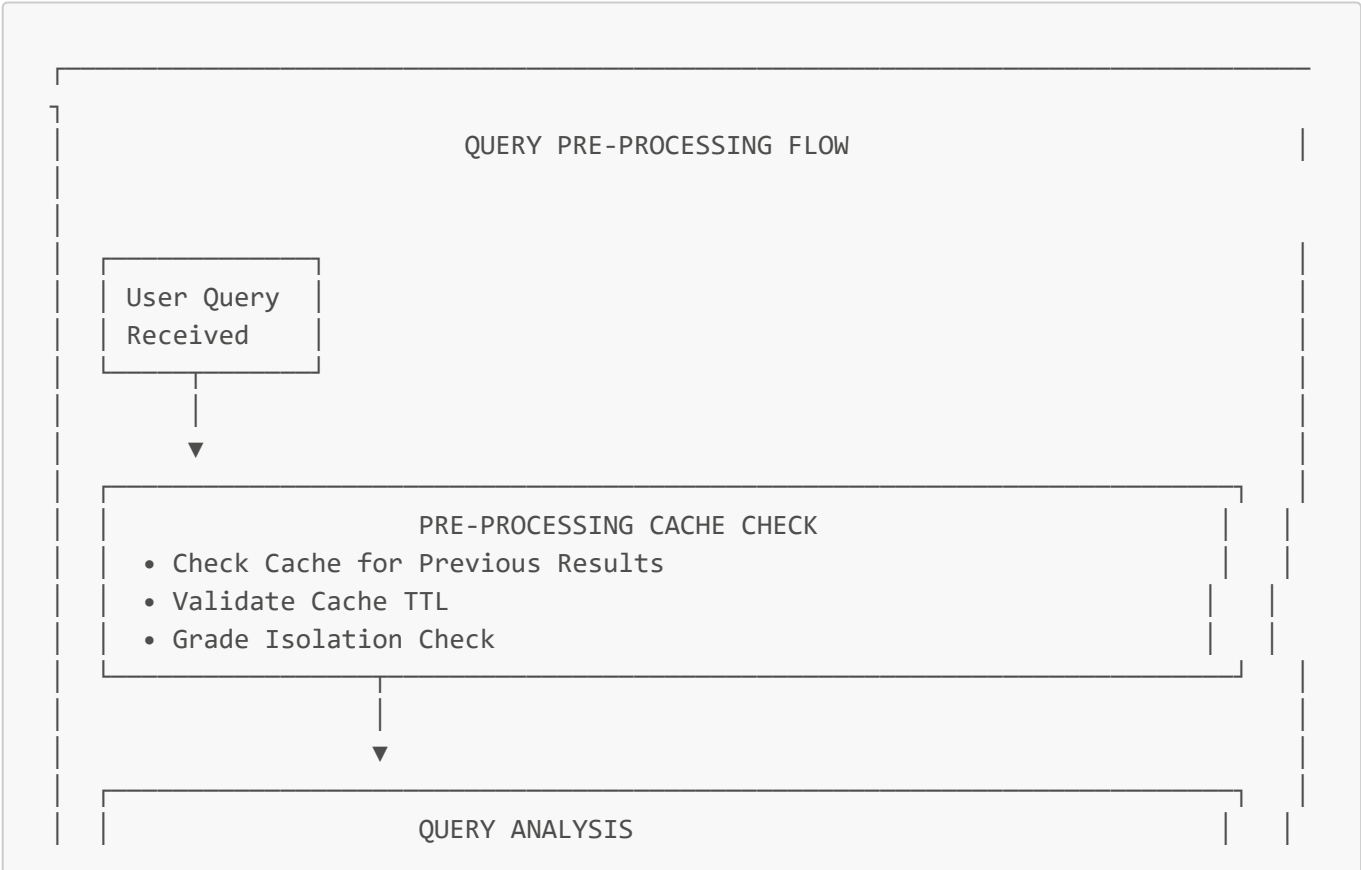


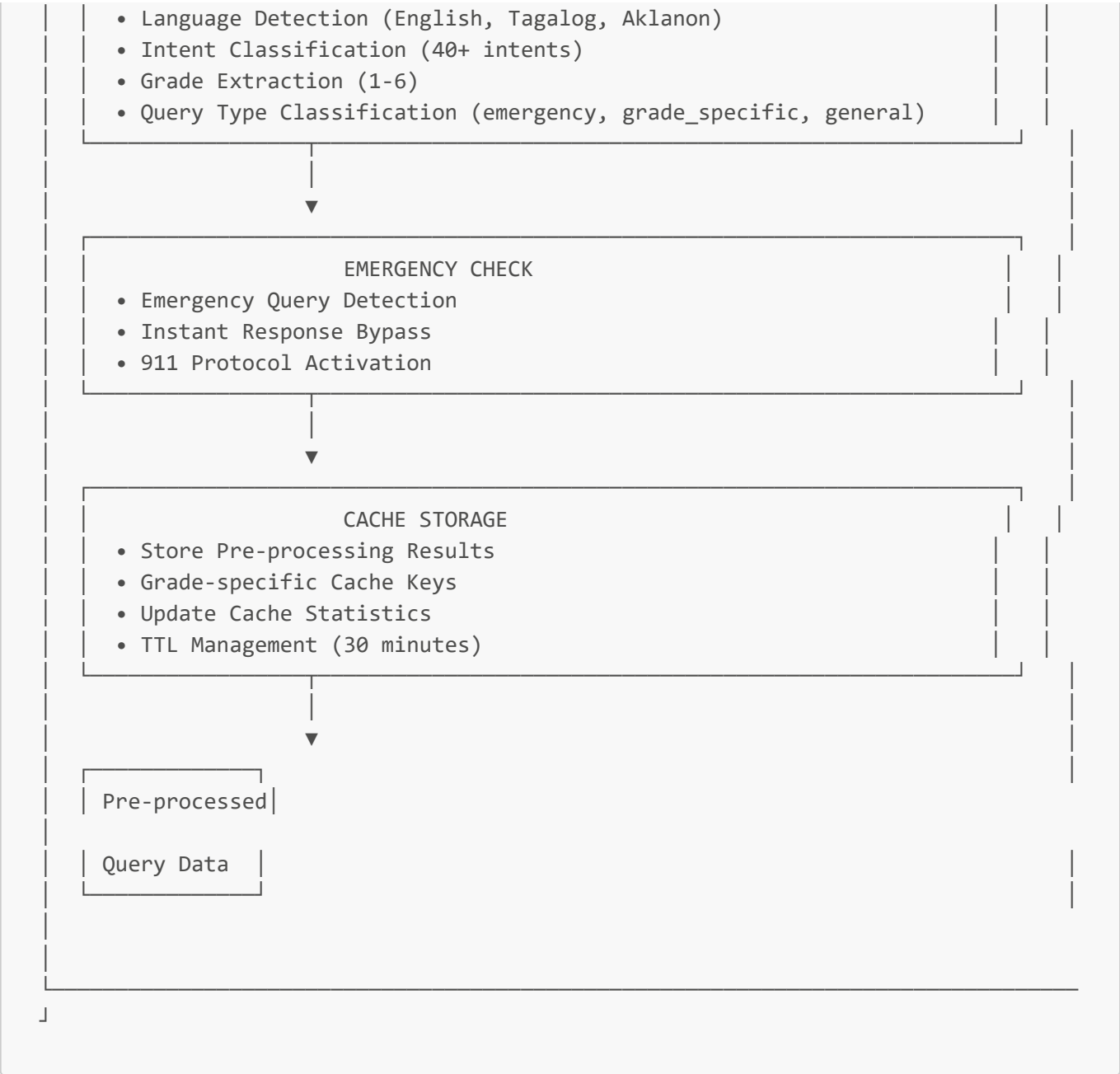
Connection Pooling Flow





Query Pre-processing Flow





Key Features Highlighted in Flowcharts

Security & Validation

- Enhanced security validation with 10+ threat types
- SQL injection protection (59+ patterns)
- XSS, command injection, and path traversal prevention
- Emergency detection with context analysis

Performance & Caching

- Redis caching at multiple stages (NLU, Database, Translation, Pre-processing)
- Intelligent cache hit/miss handling
- Fallback to in-memory cache when Redis unavailable
- TTL-based cache expiration
- Connection pooling with Transaction Mode (3-5x efficiency)
- Query pre-processing cache with grade isolation

- Automatic cache management with grade-specific invalidation

AI & Intelligence

- Multi-provider AI with intelligent fallback
- Rate limit management and provider health monitoring
- Emotional intelligence and response personalization
- Context-aware translation and cultural adaptation

Processing Capabilities

- Multi-question detection and processing (2-5 questions)
- Entity extraction and relationship detection
- Conversation memory and context preservation
- Emergency detection and crisis response
- Query pre-processing with grade isolation
- FTS search with number preservation (Grade 1-6)
- Three-tier search strategy (FTS+BM25+Fuzzy)

Error Handling & Fallbacks

- Graceful degradation when providers fail
- Multiple fallback layers (Redis → In-Memory → Direct)
- Context-aware keyword responses as final fallback
- Comprehensive error recovery mechanisms
- Connection pool fallback (Transaction → Session Mode)
- Grade-specific cache invalidation and recovery
- FTS search contamination prevention and fixes

These flowcharts provide a comprehensive visual representation of how the TOMAS chatbot processes user input, handles emergencies, manages caching, and ensures reliable response generation through multiple fallback systems. The latest improvements include database connection pooling, query pre-processing cache, and FTS search contamination fixes that significantly enhance performance and reliability.